



Yakeen Plus (2025)

PRACTICE TEST - 12

DURATION : 200 Minutes

DATE : 04/12/2024

M. MARKS : 720

Topics Covered

Physics :	Electrostatics, Current Electricity, Magnetic Effects of Current and Magnetism, Electromagnetic Induction and Alternating Currents, Electromagnetic Waves
Chemistry :	Thermodynamics & Thermochemistry, Chemical Kinetics, Alcohols, Ethers and Phenols, Aldehydes, Ketones and Carboxylic Acids, The p-Block Elements (Group 13, 14, 15, 16, 17, 18), The d- and f-Block Elements
Biology :	(Botany): Sexual Reproduction in Flowering Plant, Principle of Inheritance and Variation, Microbes in Human Welfare (Zoology): Human Reproduction, Reproductive Health, Evolution

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 20 min.** duration.
3. The test booklet consists of **200** questions. The maximum marks are **720**.
4. There are four Section in the Question Paper, Section I, II, III & IV consisting of Section-I (**Physics**), Section-II (**Chemistry**), Section-III (**Botany**) & Section IV (**Zoology**) and having **50 Questions** in each Subject and each subject is divided in two Section, **Section A** consisting 35 questions (all questions are compulsory) and **Section B** consisting 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

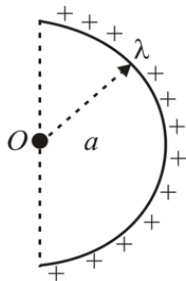
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Candidate's Signature : _____ **Invigilator's Signature** _____

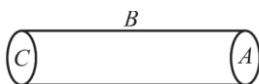
SECTION-(I) PHYSICS

SECTION - A

1. An infinite number of charges, each of charge $1 \mu\text{C}$, are placed on the x -axis with co-ordinates $x = 1, 2, 4, 8 \dots \infty$. If a charge of 1 C is kept at the origin, then what is the net force acting on 1 C charge (x is in meter).
- (1) 9000 N
 (2) 12000 N
 (3) 24000 N
 (4) 360 N
2. Electric field at the centre 'O' of a semicircle of radius ' a ' having linear charge density λ is given as:



- (1) $\frac{2\lambda}{\epsilon_0 a}$ (2) $\frac{\lambda\pi}{\epsilon_0 a}$
 (3) $\frac{\lambda}{2\pi\epsilon_0 a}$ (4) $\frac{\lambda}{\pi\epsilon_0 a}$
3. Two-point charges A and B , having charges $+Q$ and $-Q$ respectively, are placed at certain distance apart and force acting between them is F . If 25% charge of A is transferred to B , then force between the charges becomes:
- (1) F (2) $\frac{9F}{16}$
 (3) $\frac{16F}{9}$ (4) $\frac{4F}{3}$
4. A hollow cylinder has a charge q coulomb within it. If ϕ is the electric flux in units of volt meter associated with the curved surface B , the flux linked with the plane surface A in units of volt meter will be:



- (1) $\frac{q}{2\epsilon_0}$ (2) $\frac{\phi}{3}$
 (3) $\frac{q}{\epsilon_0} - \phi$ (4) $\frac{1}{2} \left(\frac{q}{\epsilon_0} - \phi \right)$

5. Acceleration of a charged particle of charge ' q ' and mass ' m ' moving in a uniform electric field of strength ' E ' is:

- (1) $\frac{qE}{m}$ (2) $\frac{m}{qE}$
 (3) mqE (4) $\frac{q}{mE}$

6. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: An electrostatic field line never forms a closed loop.

Reason R: Electrostatic field is a conservative field.

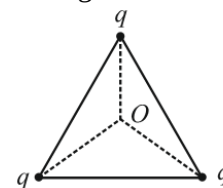
In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is not the correct explanation of A.

7. 1000 similar electrified rain drops merge together into one drop so that their total charge remains unchanged. The electric energy gets increased by:

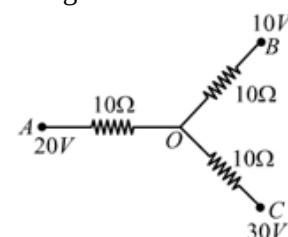
- (1) 100 times. (2) 200 times.
 (3) 300 times. (4) 400 times.

8. Three isolated equal charges are placed at the three corners of an equilateral triangle as shown in figure. The statement which is true for net electric potential V and net electric field intensity E at the centre of the triangle is:



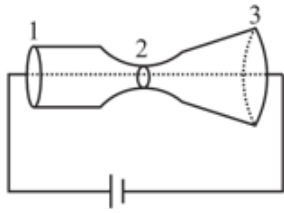
- (1) $E = 0, V = 0$ (2) $V = 0, E \neq 0$
 (3) $V \neq 0, E = 0$ (4) $V \neq 0, E \neq 0$

9. Current through the branch AO is:

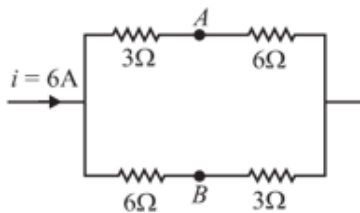


- (1) 2 A (2) 4 A
 (3) 1 A (4) Zero

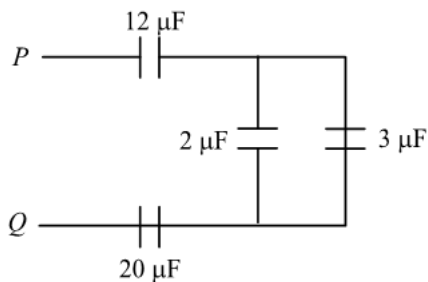
10. For given conductor the drift speed of electrons at cross section (1), (2) and (3) are v_1 , v_2 and v_3 , then:



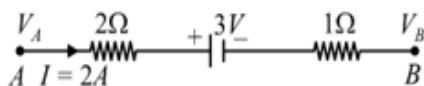
- (1) $v_1 = v_2 = v_3$
 (2) $v_1 < v_2 < v_3$
 (3) $v_2 > v_1 > v_3$
 (4) $v_2 > v_3 > v_1$
11. The potential difference between points A and B is equal to:



- (1) 6 V
 (2) 9 V
 (3) -6 V
 (4) 12 V
12. In the circuit diagram shown in the adjoining figure, the resultant capacitance between P and Q is:

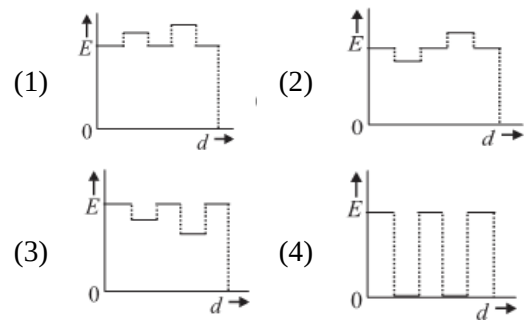
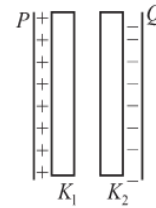


- (1) $47\mu F$
 (2) $3\mu F$
 (3) $60\mu F$
 (4) $10\mu F$
13. The potential difference ($V_A - V_B$) between the points A and B in the given figure is:



- (1) -3V
 (2) +3V
 (3) +6V
 (4) +9V

14. Two thin dielectric slabs of dielectric constants K_1 and K_2 ($K_1 < K_2$) are inserted between plates of a parallel plate capacitor, as shown in the figure. The variation of electric field E between the plates with distance d as measured from plate P is correctly shown by:



15. Three equal capacitors each with capacitance C are connected as shown in figure then the equivalent capacitance is:



- (1) C
 (2) $3C$
 (3) $C/3$
 (4) $3C/2$
16. Two charged particles having same linear momentum describe circular paths of radius R_1 and R_2 respectively in uniform transverse magnetic field. If the ratio of their respective charges is, then the ratio of R_1 and R_2 will be:

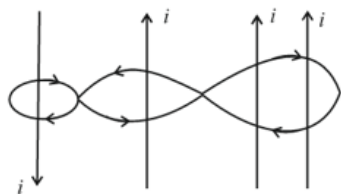
- (1) 1 : 3
 (2) 3 : 1
 (3) 1 : 9
 (4) 9 : 1
17. The value of $\oint \vec{B} \cdot d\vec{l}$ for the loop as shown in the figure will be:



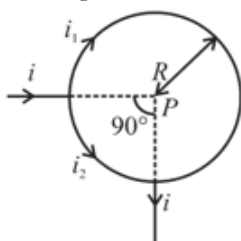
- (1) $2\mu_0 i$
 (2) $3\mu_0 i$
 (3) $\mu_0 i$
 (4) $5\mu_0 i$
18. When a charged particle enters a uniform magnetic field?

- (1) Speed of particle must increase
 (2) Speed of particle must decrease
 (3) Speed of particle will be constant
 (4) Speed of particle may be decrease or increase

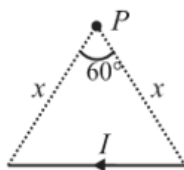
19. Figure shows four identical currents and an amperian loops encircling them. We shall calculate $\oint \vec{B} \cdot d\vec{l}$ in the direction marked. The **correct** value is:



- (1) $\oint \vec{B} \cdot d\vec{l} = 2\mu_0 i$ (2) $\oint \vec{B} \cdot d\vec{l} = 0$
 (3) $\oint \vec{B} \cdot d\vec{l} = \mu_0 i$ (4) $\oint \vec{B} \cdot d\vec{l} = 4\mu_0 i$
20. A straight conductor carrying current i splits into two parts as shown in the figure. The radius of the circular loop is R . The total magnetic field at the centre P of the loop is:

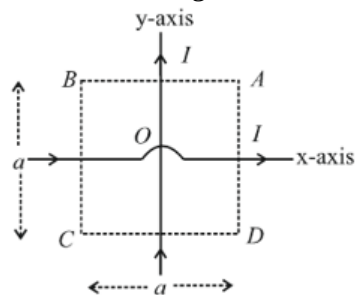


- (1) $\frac{\mu_0 i}{2R}$, inward
 (2) Zero
 (3) $3\mu_0 i / 32R$, outward
 (4) $3\mu_0 i / 32R$, inward
21. A long wire carrying a steady current is bent into a circular loop of one turn. The magnetic field at the centre of the loop is B . It is then bent into a circular coil of n turns. The magnetic field at the centre of this coil of n turns will be;
- (1) nB
 (2) n^2B
 (3) $2nB$
 (4) $2n^2B$
22. A straight wire of finite length carrying current I subtends an angle of 60° at point P as shown. The magnetic field at P is:



- (1) $\frac{\mu_0 I}{2\sqrt{3}\pi x}$ (2) $\frac{\mu_0 I}{2\pi x}$
 (3) $\frac{\sqrt{3}\mu_0 I}{2\pi x}$ (4) $\frac{\mu_0 I}{3\sqrt{3}\pi x}$

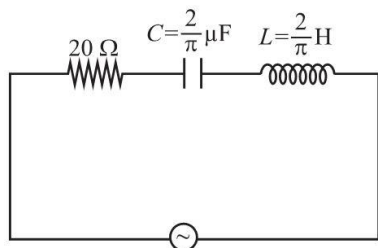
23. Two infinite length wires carry equal current and placed along x and y axis respectively. At which points the resultant magnetic field is zero?



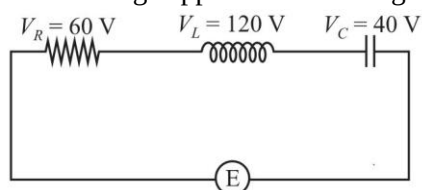
- (1) A, B (2) B, D
 (3) A, C (4) C, D
24. Flux ϕ (in weber) in a closed circuit of resistance 10 ohm varies with time t (in second) according to the equation $\phi = 6t^2 - 5t + 1$. What is the magnitude of the induced current at $t = 0.25$ s?
- (1) 1.2 A
 (2) 0.2 A
 (3) 0.6 A
 (4) 0.8 A
25. According to Faraday's law of electromagnetic induction:
- (1) electric field is produced by time varying magnetic flux.
 (2) magnetic field is produced by time varying electric flux.
 (3) magnetic field is associated with a moving charge.
 (4) None of these
26. In which type of material the magnetic susceptibility does **not** depend on temperature?
- (1) Diamagnetic
 (2) Paramagnetic
 (3) Ferromagnetic
 (4) Ferrite
27. The magnetic flux through a circuit of resistance R changes by an amount $\Delta \phi$ in a time Δt . Then the total quantity of electric charge q that passes any point in the circuit during the time Δt is represented by:

- (1) $q = \frac{1}{R} \frac{\Delta \phi}{\Delta t}$
 (2) $q = \frac{\Delta \phi}{\Delta t}$
 (3) $q = \frac{\Delta \phi}{\Delta t}$
 (4) $q = R \cdot \frac{\Delta \phi}{\Delta t}$

28. In an AC circuit as shown in the figure, the source is of r.m.s voltage 200 V and variable frequency. At resonance, the circuit:



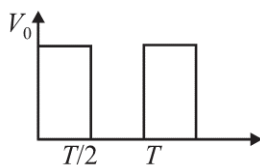
- (1) has a r.m.s current 10 A.
 (2) has a resonant frequency 250 Hz.
 (3) has voltage across capacitor, inductor and resistor in same phase.
 (4) Both (1) and (2)
29. What is the voltage applied in following circuit?



- (1) 220 V
 (2) 140 V
 (3) 120 V
 (4) 100 V
30. When LCR series circuit is in state of resonance, its resistance is doubled, then:
- (1) Q -factor of circuit will increase.
 (2) Q -factor of circuit will decrease.
 (3) Q -factor of circuit remains unchanged.
 (4) frequency of resonance will be halved.
31. The quality factor of series $L-C-R$ circuit is Q , then the value capacitance in terms of resistance R and inductance (L) is:

(1) $\frac{L^2}{R^2 Q^2}$ (2) $C = \frac{L}{(RQ)^2}$
 (3) $\frac{R^2 Q^2}{L^2}$ (4) $\frac{(RQ)^2}{L}$

32. The rms value of potential difference V shown in the figure is:



(1) $\frac{V_0}{2}$ (2) $\frac{V_0}{\sqrt{3}}$
 (3) V_0 (4) $\frac{V_0}{\sqrt{2}}$

33. A coil has resistances 30Ω and inductive reactance 20Ω at 50 Hz frequency. If an ac source, of 200 V, 100 Hz is connected across the coil, the current in the coil will be:

(1) $\frac{20}{\sqrt{13}} \text{ A}$ (2) 2.0 A
 (3) 4.0 A (4) 8.0 A

34. An alternating current is given by the equation $i = i_1 \sin \omega t + i_2 \cos \omega t$. The rms current will be:

(1) $\frac{1}{2}(i_1^2 + i_2^2)^{1/2}$ (2) $\frac{1}{\sqrt{2}}(i_1^2 + i_2^2)^{1/2}$
 (3) $\frac{1}{\sqrt{2}}(i_1 + i_2)^2$ (4) $\frac{1}{\sqrt{2}}(i_1 + i_2)$

35. In a LCR circuit having $L = 8.0$ henry, $C = 0.5 \mu\text{F}$ and $R = 100$ ohm in series. The resonance frequency in per second is:

(1) 600 radian (2) 600 Hz
 (3) 500 radian (4) 500 Hz

SECTION - B

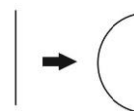
36. In an oscillating LC circuit the maximum charge on the capacitor is Q . The charge on the capacitor when the energy is stored equally between the electric and magnetic fields is:

(1) $Q/2$ (2) $Q/\sqrt{3}$
 (3) $Q/\sqrt{2}$ (4) Q

37. A $60 \mu\text{F}$ capacitor is charged to 100 volts. This charged capacitor is connected across a 1.5 mH coil, so that LC oscillations occur. The maximum current in the coil is:

(1) 1.5 A (2) 2 A
 (3) 15 A (4) 20 A

38. A magnetised wire of moment M is bent into an arc of a circle subtending an angle of 60° at the centre. The new magnetic moment is:

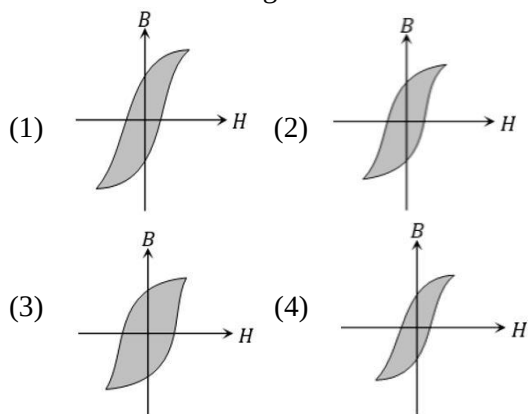


(1) $(2M/\pi)$ (2) (M/π)
 (3) $(3\sqrt{3}M/\pi)$ (4) $(3M/\pi)$

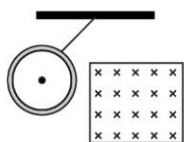
39. A magnet of magnetic moment $50\hat{i}$ A-m² is placed along the x -axis in a magnetic field $\vec{B} = (0.5\hat{i} + 3.0\hat{j})$ T. The torque acting on the magnet is:

(1) $175 \hat{k}$ N-m (2) $150 \hat{k}$ N-m
 (3) $75 \hat{k}$ N-m (4) $25\sqrt{37} \hat{k}$ N-m

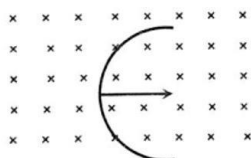
40. For substances hysteresis ($B - H$) curves are given as shown in figure. For making temporary magnet which of the following is the best?



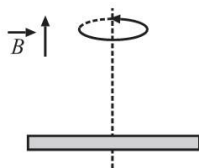
41. A metallic ring connected to a rod oscillates freely like a pendulum. If now a magnetic field is applied in horizontal direction so that the pendulum now swings through the field, the pendulum will:



- (1) keep oscillating with the old time period.
 (2) keep oscillating with a smaller time period.
 (3) keep oscillating with a larger time period.
 (4) come to rest very soon.
42. A straight wire of length L is bent into a semicircle. It is moved in a uniform magnetic field with speed v with diameter perpendicular to the field. The induced emf between the ends of the wire is:

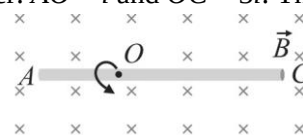


- (1) BLv (2) $2BLv$
 (3) $2\pi BLv$ (4) $\frac{2BvL}{\pi}$
43. A conducting rod of length $2l$ is rotating with constant angular speed ω about its perpendicular bisector. A uniform magnetic field $\vec{B}$ exists parallel to the axis of rotation. The e.m.f. induced between two ends of the rod is:



- (1) $B\omega l^2$ (2) $\frac{1}{2}B\omega l^2$
 (3) $\frac{1}{8}B\omega l^2$ (4) Zero

44. A conducting rod AC of length $4l$ is rotated about a point O in a uniform magnetic field $\vec{B}$ directed into the paper. $AO = l$ and $OC = 3l$. Then:



- (1) $V_A - V_O = \frac{B\omega l^2}{2}$ (2) $V_O - V_C = \frac{7}{2}B\omega l^2$
 (3) $V_A - V_C = 4B\omega l^2$ (4) $V_C - V_O = \frac{9}{2}B\omega l^2$

45. Two identical short bar magnets, each having magnetic moment M , are placed a distance of $2d$ apart with axes perpendicular to each other in a horizontal plane. The magnetic induction at a point midway between them is:

- (1) $\frac{\mu_0}{4\pi}(\sqrt{2})\frac{M}{d^3}$ (2) $\frac{\mu_0}{4\pi}(\sqrt{3})\frac{M}{d^3}$
 (3) $\left(\frac{2\mu_0}{\pi}\right)\frac{M}{d^3}$ (4) $\frac{\mu_0}{4\pi}(\sqrt{5})\frac{M}{d^3}$

46. The speed of electromagnetic waves depends upon:

- (1) Wavelength
 (2) Frequency
 (3) Intensity
 (4) Medium, in which it travels

47. Out of the following options which one can be used to produce a propagating electromagnetic wave?

- (1) A stationary charge.
 (2) A chargeless particle.
 (3) An accelerating charge.
 (4) A charge moving at constant velocity.

48. A plane electromagnetic wave of frequency 20 MHz travels through a space along x -direction. If the electric field vector at a certain point in space is 6 Vm^{-1} , then what is the magnetic field vector at that point?

- (1) $2 \times 10^{-8} \text{ T}$ (2) $1/2 \times 10^{-8} \text{ T}$
 (3) 2 T (4) $1/2 \text{ T}$

49. The electromagnetic wave with shortest wavelength among the following is:

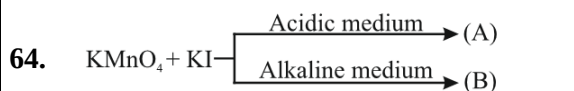
- (1) UV-rays (2) X-rays
 (3) γ -rays (4) microwaves

50. In electromagnetic wave, according to Maxwell, changing electric field gives:

- (1) stationary magnetic field
 (2) conduction current
 (3) eddy current
 (4) displacement current

SECTION-(II) CHEMISTRY**SECTION - A**

51. When a substance is heated, its entropy increases. The increase in entropy will be maximum at:
 (1) 0°C (2) the melting point
 (3) the boiling point (4) 100°C
52. Which of the following factors affect the internal energy of the system? When:
 (1) heat passes into or out of the system.
 (2) work is done on or by the system.
 (3) matter enters or leaves the system.
 (4) all of the above
53. The entropy change (J/K) involved in the isothermal reversible expansion of 2 moles of an ideal gas from a volume of 10 dm³ to a volume of 100 dm³ at 27°C is:
 (1) 22.19 (2) 89.28
 (3) 38.29 (4) 18.58
54. The bond dissociation energies of X₂, Y₂ and XY are in the ratio of 1 : 0.5 : 1, ΔH for the formation of XY is -200 kJ mol⁻¹. The bond dissociation energy of X₂ will be:
 (1) 200 kJ mol⁻¹ (2) 100 kJ mol⁻¹
 (3) 400 kJ mol⁻¹ (4) 800 kJ mol⁻¹
55. A reversible isothermal evaporation of 90g of water is carried out at 100°C. Heat of evaporation of water is 9.72 kCal/mol. Assuming water vapour to behave like an ideal gas. What is the change in internal energy of the system?
 (1) 48.6 kCal (2) 52.33 kCal
 (3) 44.87 kCal (4) 56.06 kCal
56. The number of unpaired electron in gaseous species of Mn³⁺, Cr³⁺ and V³⁺ respectively are:
 (1) 4, 2 and 2 (2) 3, 3 and 2
 (3) 4, 3 and 2 (4) 3, 3 and 3
57. Which of the following is **incorrect** about interstitial compounds made by d-block elements?
 (1) These compounds are usually non-stoichiometric.
 (2) These compounds retain metallic conductivity.
 (3) These compounds have melting point greater than those of pure metals.
 (4) Atoms trapped inside the lattice make the compound highly reactive.
58. If the rate of reaction increases to two times with every 10°C rise in temperature, then if we raise the temperature by 40°C, the new rate of the reaction will be:
 (1) 4 times
 (2) 8 times
 (3) 16 times
 (4) 32 times
59. The rate constant k₁ and k₂ for two different reactions are 10¹⁶.e^{-2000/T} and 10¹⁵.e^{-1000/T}, respectively. The temperature at which k₁ = k₂ is:
 (1) 1000 K (2) $\frac{2000}{2.303}$ K
 (3) 2000 K (4) $\frac{1000}{2.303}$ K
60. According to collision theory, most molecular collisions don't lead to reaction. Which of the following are necessary for collisions to successfully lead to reaction?
 A. The colliding particles must have sufficient kinetic energy (threshold energy)
 B. A catalyst must be present at the collision.
 C. The colliding particles must be properly oriented in space when collide.
 Choose the **correct** answer from the options given below:
 (1) A, B and C
 (2) A only
 (3) A and C only
 (4) B and C only
61. 2A + B → C + D
 In this reaction, if we double the concentration of A, reaction rate become two times. And in other experiment, we double the concentration of A and B, reaction rate again become two times. What is the order of this reaction?
 (1) 1 (2) 3
 (3) 2 (4) 1.5
62. At 298 K if 60% of first order reaction was completed in 60 min, 50%, of the same reaction would be completed in approximately: (log 2.5 = 0.3979)
 (1) 45 min (2) 60 min
 (3) 40 min (4) 50 min
63. Formula of halide of lanthanoid may be:
 (1) LnX₃ (2) LnX₅
 (3) LnX₂ (4) LnX₆



The product (A) and (B) are respectively:

- (1) KI and I_2
- (2) I_2 and KIO_3
- (3) KIO_3 and KIO_4
- (4) I_2 and I_2

65. Given below are two statements:

Statement I: The actinoids show in general +3 oxidation state.

Statement II: Actinoid contraction is greater from element to element than lanthanoid contraction.

In the light of the above statements, choose the most appropriate answer from the options given below:

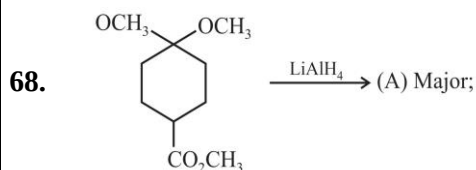
- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

66. The first ionisation enthalpy of the lanthanoids are comparable to those of:

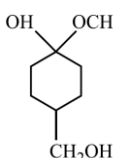
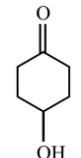
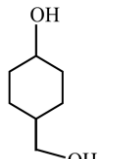
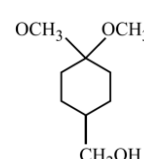
- (1) Sodium
- (2) Calcium
- (3) Silicon
- (4) Magnesium

67. Which of the following alkanols is most soluble in water?

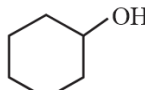
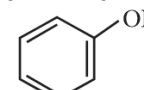
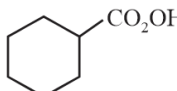
- (1) 1-Butanol
- (2) 2-Butanol
- (3) Isobutyl Alcohol
- (4) t-Butyl Alcohol



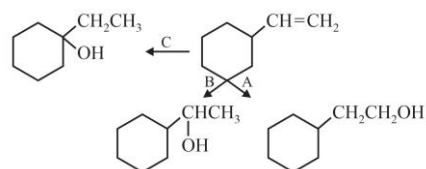
product (A) is:

- (1) 
- (2) 
- (3) 
- (4) 

69. The structural formula of cyclohexyl alcohol is:

- (1) 
- (2) 
- (3) 
- (4) None of these

70. Select schemes A, B, C out of



- I. Acid catalyzed hydration
- II. Hydroboration-oxidation
- III. Oxymercuration-demercuration
- (1) I in all cases
- (2) I, II, III
- (3) II, III, I
- (4) III, I, II

71. Alkyl halide on heating with dry Ag_2O produces:

- (1) Ether
- (2) Ester
- (3) Ketone
- (4) Hydrocarbon

72. **Correct** order of reactivity of CH_3CHO , $\text{C}_2\text{H}_5\text{COCH}_3$ and CH_3COCH_3 with HCN is:

- (1) $\text{CH}_3\text{CHO} > \text{CH}_3\text{COCH}_3 > \text{CH}_3\text{COC}_2\text{H}_5$
- (2) $\text{C}_2\text{H}_5\text{COCH}_3 > \text{CH}_3\text{COCH}_3 > \text{CH}_3\text{CHO}$
- (3) $\text{CH}_3\text{COCH}_3 > \text{CH}_3\text{CHO} > \text{C}_2\text{H}_5\text{COCH}_3$
- (4) $\text{CH}_3\text{COCH}_3 > \text{C}_2\text{H}_5\text{COCH}_3 > \text{CH}_3\text{CHO}$

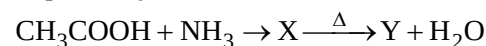
73. Which one of the compound is **not** known?

- (1) NCl_5
- (2) PCl_5
- (3) AsCl_5
- (4) NCl_3

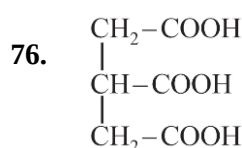
74. How many bridged oxygen atoms are present in chromate and dichromate anions, respectively?

- (1) Zero and two
- (2) Zero and one
- (3) One and zero
- (4) Two and one

75. In the following reaction, X and Y are respectively :



- (1) $\text{CH}_3\text{CONH}_2, \text{CH}_4$
- (2) $\text{CH}_3\text{COO}^-\text{NH}_4^+, \text{CH}_3\text{CONH}_2$
- (3) $\text{CH}_3\text{CONH}_2, \text{CH}_3\text{COOH}$
- (4) $\text{CH}_3\text{NH}_2, \text{CH}_3\text{COOH}_2$



IUPAC name is:

- (1) Propane-1, 2, 3-tricarboxylic acid
- (2) 2-Carboxy pentane dioic acid
- (3) Propane tricarboxylic acid
- (4) Propane-1, 2, 3-tricarboxylic acid

77. Four compounds A, B, C and D were heated individually with Tollen's reagent. It was found that all compounds other than D formed a silver mirror on the inside of their test tubes. Identify the compound D:

- (1) Acetaldehyde
- (2) Butyraldehyde
- (3) Acetone
- (4) Benzaldehyde

78. Which among the following will give positive 2, 4-DNP test?

- (1) C_6H_5OH
- (2) $(CH_3)_3COH$
- (3) $CH_3OC_2H_5$
- (4) C_2H_5CHO

79. Volatile nature of halogens is because:

- (1) Halogen molecules are bonded by strong forces.
- (2) Halogen molecules are bonded by electrostatic forces.
- (3) The forces existing between the discrete molecules are only weak van der Waals forces.
- (4) Halogen molecules are more reactive.

80. Which of the following hydride is most stable?

- (1) AsH_3
- (2) SbH_3
- (3) PH_3
- (4) NH_3

81. The increasing order of atomic radii of the following group 13 elements is:

- (1) $Al > Ga < In < Tl$
- (2) $Ga < Al < In < Tl$
- (3) $Al < In < Ga < Tl$
- (4) $Al < Ga < Tl < In$

82. Between which of the following elements is the discontinuity in the ionisation enthalpy values observed?

- (1) Al and Ga
- (2) In and Tl
- (3) B and Al
- (4) Both (1) and (2)

83. **Correct** order of reducing capacity is:

- (1) $Ge^{2+} > Sn^{2+} > Pb^{2+}$
- (2) $Ge^{2+} < Sn^{2+} < Pb^{2+}$
- (3) $Ge^{2+} \approx Sn^{2+} \approx Pb^{2+}$
- (4) $Pb^{2+} > Ge^{2+} > Sn^{2+}$

84. Carbon-60 contains:

- (1) 20 pentagons and 12 hexagons
- (2) 12 pentagons and 20 hexagons
- (3) 30 pentagons and 30 hexagons
- (4) 24 pentagons and 36 hexagons

85. Boron compounds behave as Lewis acids because of their:

- (1) Covalent nature
- (2) Electron deficient character
- (3) Ionization property
- (4) Ionic nature

SECTION - B

86. Among H_2O , H_2S , H_2Se and H_2Te the one with maximum boiling point is:

- (1) H_2O due to H-bonding.
- (2) H_2Te due to higher molecular mass.
- (3) H_2S due to H-bonding.
- (4) H_2Se due to lower molecular mass.

87. Element having highest catenation property is:

- (1) S
- (2) O
- (3) Se
- (4) Te

88. The three important oxidation states of phosphorus are:

- (1) $-3, +3$ and $+5$
- (2) $-3, +3$ and -5
- (3) $-3, +3$ and $+2$
- (4) $-3, +3$ and $+4$

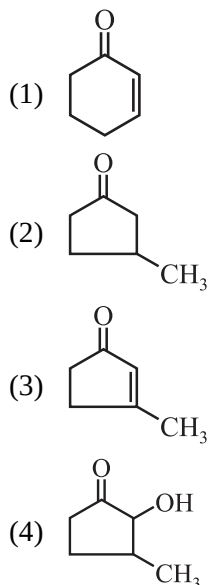
89. Elements of group 14:

- (1) exhibit oxidation state of $+4$ only
- (2) exhibit oxidation state of $+2$ and $+4$
- (3) exhibit oxidation state of $+1$ and $+3$
- (4) exhibit oxidation state of $+3$ and $+5$

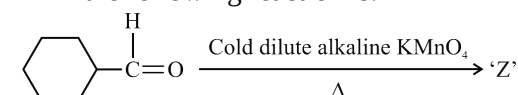
90. The **correct** increasing order of the stability of Al^+ , Ga^+ , In^+ , Tl^+ ions are:

- (1) $Al^+ < Ga^+ < Tl^+ < In^+$
- (2) $Al^+ < Ga^+ < In^+ < Tl^+$
- (3) $Tl^+ < In^+ < Ga^+ < Al^+$
- (4) $Tl^+ < Al^+ < Ga^+ < In^+$

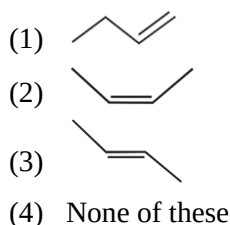
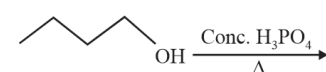
91. The diketone $\text{CH}_3-\text{C}(=\text{O})-(\text{CH}_2)_2-\text{C}(=\text{O})-\text{CH}_3$ on base catalysed condensation gives the final product as:



92. 'Z' in the following reaction is:



- (1) cyclohexanone
 (2) cyclohexane
 (3) benzophenone
 (4) cyclohexane carboxylic acid
93. For a zero-order reaction, the plot of conc. $(a - x)$ vs time is linear with:
- (1) +ve slope and zero intercept
 (2) -ve slope and zero intercept
 (3) +ve slope and non-zero intercept
 (4) -ve slope and non-zero intercept
94. All radioactive disintegration reactions are examples of:
- (1) second order kinetics
 (2) zero order kinetics
 (3) first order kinetics
 (4) third order kinetics
95. Find out the major product of the following reaction:



96. $\text{C}_2\text{H}_5\text{OH}$ reacts with halogen acids in the presence of ZnCl_2 to form $\text{C}_2\text{H}_5\text{X}$. Its reactivity with HI, HBr and HCl follows the order:

- (1) $\text{HI} > \text{HBr} > \text{HCl}$
 (2) $\text{HCl} > \text{HBr} > \text{HI}$
 (3) $\text{HCl} > \text{HI} > \text{HBr}$
 (4) $\text{HBr} > \text{HCl} > \text{HI}$

97. P and Q are arbitrarily chosen intensive variables then:

- A. $(P + Q)$ is extensive property
 B. P/Q is an intensive property
 C. PQ is an intensive property
 D. dP/dQ is an intensive property

Choose the **correct** answer from the options given below:

- (1) A and B only
 (2) B and C only
 (3) B and D only
 (4) B, C and D only

98. The enthalpy of vaporization of chloroform is 35 kJ mol^{-1} at its normal boiling point of 77°C . What is the entropy of condensation of chloroform at this temperature?

- (1) $-90 \text{ J mol}^{-1} \text{ K}^{-1}$
 (2) $-100 \text{ J mol}^{-1} \text{ K}^{-1}$
 (3) $-110 \text{ J mol}^{-1} \text{ K}^{-1}$
 (4) $-200 \text{ J mol}^{-1} \text{ K}^{-1}$

99. Among the following series of transition metal ions, the one in which all metal ions have $3d^2$ electronic configuration is:

- (1) $\text{Ti}^{4+}, \text{V}^{3+}, \text{Cr}^{2+}, \text{Mn}^{3+}$
 (2) $\text{Ti}^{2+}, \text{V}^{3+}, \text{Cr}^{4+}, \text{Mn}^{5+}$
 (3) $\text{Ti}^{3+}, \text{V}^{2+}, \text{Cr}^{3+}, \text{Mn}^{4+}$
 (4) $\text{Ti}^+, \text{V}^{4+}, \text{Cr}^{6+}, \text{Mn}^{7+}$

100. The shape of XeOF_2 on the basis of VSEPR theory is:

- (1) see-saw.
 (2) V-shape.
 (3) trigonal planar.
 (4) T-shape.

SECTION-(III) BOTANY**SECTION - A**

101. Which of the following is **not** a Mendelian disorder?

- (1) Cystic fibrosis
- (2) Thalassemia
- (3) Haemophilia
- (4) Turner's Syndrome

102. Select the statement which is **incorrect**?

- (1) Polygenic character is controlled by multiple alleles
- (2) In case of polygenic inheritance, thousands of intermediate phenotypes are found between two extreme ones
- (3) Height, weight, skin colour are polygenic
- (4) Polygenic character is controlled by multiple genes

103. If a heterozygous axial flower producing pea plant is crossed with plant with terminal flower, what percent of offspring will be with axial flower?

- (1) 25%
- (2) 50%
- (3) 75%
- (4) 100%

104. If a genetic disease is transferred from a phenotypically normal but carrier female to only some of the male progeny, the disease is:

- (1) autosomal dominant.
- (2) autosomal recessive.
- (3) sex-linked dominant.
- (4) sex-linked recessive.

105. In a test cross of dominant phenotype pea plant having violet colour flower with recessive parent result in violet and white flower, then genotype for test cross is:

- (1) $Ww \times Ww$.
- (2) $Tt \times tt$
- (3) $Ww \times ww$.
- (4) $WW \times ww$.

106. Select the **incorrect** statement w.r.t. phenylketonuria.

- (1) It is an autosomal recessive trait.
- (2) The affected individuals lack enzyme phenylalanine hydroxylase.
- (3) In the affected individual, phenylalanine is converted into tyrosine.
- (4) Affected individuals experience mental retardation.

107. Match List-I with List-II.

List-I (Characters)		List-II (Contrasting traits)	
(A)	Stem height	(I)	Green
(B)	Flower colour	(II)	Constricted
(C)	Pod shape	(III)	Dwarf
(D)	Seed colour	(IV)	Violet

Choose the **correct** answer from the options

- (1) A-III, B-IV, C-II, D-I
- (2) A-II, B-IV, C-III, D-I
- (3) A-III, B-I, C-II, D-IV
- (4) A-I, B-II, C-III, D-IV

108. A gene showing co-dominance has:

- (1) alleles that are recessive to each other.
- (2) both alleles independently expressed in the heterozygote.
- (3) one allele dominant on the other.
- (4) alleles tightly linked on the same chromosome.

109. Identify the **wrongly** matched pair.

- (1) Klinefelter's syndrome - Sterility in males
- (2) Turner's syndrome - XXY condition
- (3) Haemophilia - Sex linked recessive disease
- (4) Thalassemia - Anaemia

110. A marriage between normal visioned man and colourblind woman will produce:

- (1) colourblind sons and 50% carrier daughters.
- (2) 50% colourblind sons and 50% carrier daughters.
- (3) normal sons and carrier daughters.
- (4) colourblind sons and carrier daughters.

111. If two genes experience independent assortment, which assumption is most likely true?

- (1) They are located in close proximity on the same chromosome
- (2) Crossing over between the genes does not occur
- (3) The genes are located on different chromosomes or are far apart on the same chromosome
- (4) The expression of one gene does not affect the expression of the other

112. Choose the set of sex determining chromosomes that indicates the **correct** sex of the respective organism.
- (1) Homozygous sex chromosomes (ZZ) determine female sex in birds
 - (2) XO type of sex chromosomes determine male sex in grasshoppers
 - (3) XXY condition in humans, as found in Turner Syndrome, determines male sex
 - (4) Homozygous sex chromosomes (XX) produce male sex in *Drosophila*

113. If an individual with genotype TtRr is crossed with similar genotype and if both gene T and R are linked and crossing over not occur then probable genotype is:
- (1) TTRR, TtRR, TtRr, Ttrr.
 - (2) ttrr, ttRr, TtRr, TTRr.
 - (3) TtRr, ttrr, TTRR.
 - (4) TtRr, ttrr, TTrr.

114. Given below are two statements:
- Statement I:** ABO blood groups are controlled by the gene *I*.
- Statement II:** The plasma membrane of the red blood cells has sugar polymers that protrude from its surface and the kind of sugar is controlled by the *I* gene.
- In the light of the above statements, choose the **correct** answer from the options given below:
- (1) Statement I is correct but Statement II is incorrect.
 - (2) Statement I is incorrect but Statement II is correct.
 - (3) Both Statement I and Statement II are correct.
 - (4) Both Statement I and Statement II are incorrect.

115. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
- Assertion (A):** The Punnett square is used to understand production of gametes by the parents.
- Reason (R):** Punnett square is a graphical representation to calculate the probability of all possible genotypes of offspring in a genetic cross.
- In the light of the above statements, choose the **correct** answer from the options given below:
- (1) A is true but R is false.
 - (2) A is false but R is true.
 - (3) Both A and R are true and R is the correct explanation of A.
 - (4) Both A and R are true but R is NOT the correct explanation of A.

116. The domestic sewage in large cities:
- (1) has a high BOD as it contains both aerobic and anaerobic bacteria.
 - (2) is processed by aerobic and then anaerobic bacteria in the secondary treatment in Sewage Treatment Plants (STPs).
 - (3) when treated in STPs does not really require the aeration step as the sewage contains adequate oxygen.
 - (4) has very high amount of suspended solids and dissolved salts.

117. Match **list I** with **list II**.

List-I		List-II	
(A)	Citric acid	(I)	<i>Saccharomyces cerevisiae</i>
(B)	Baker's yeast	(II)	<i>Nucleopolyhedrovirus</i>
(C)	Statins	(III)	<i>Aspergillus niger</i>
(D)	Biocontrol agent	(IV)	<i>Monascus purpureus</i>

Choose the **correct** answer from the options given below:

- (1) A-I, B-IV, C-II, D-III
 - (2) A-III, B-IV, C-I, D-II
 - (3) A-III, B-I, C-II, D-IV
 - (4) A-III, B-I, C-IV, D-II
118. Cyclosporin A is used for:
- (1) dissolving blood clots.
 - (2) lowering cholesterol level.
 - (3) immunosuppression.
 - (4) production of citric acid.
119. One of the following is **not** a biofertiliser:
- (1) *Azotobacter*
 - (2) *Bacillus thuringiensis*
 - (3) *Anabaena*
 - (4) *Nostoc*
120. Given below are two statements:
- Statement-I:** Discovery of first antibiotic, penicillin was a chance discovery.
- Statement-II:** Alexander Fleming while working on *Streptococcus* observed a mould of *Penicillium notatum* growing in unwashed plates.
- In the light of the above statements, choose the most appropriate answer from the options given below:
- (1) Statement I is correct but Statement II is incorrect.
 - (2) Statement I is incorrect but Statement II is correct.
 - (3) Both Statement I and Statement II are correct.
 - (4) Both Statement I and Statement II are incorrect.

121. In a sewage treatment plant, primary treatment is:

- (1) physical process which involves sedimentation only.
- (2) physical process which involves both filtration and sedimentation.
- (3) biological process which involves formation of primary sludge and effluent.
- (4) biological process which involves both filtration and sedimentation.

122. Big holes in Swiss cheese are formed as a result of:

- (1) machine used for making the cheese.
- (2) a bacterium producing a large amount of carbon dioxide.
- (3) a bacterium that produces large amount of oxygen.
- (4) a fungus that produces large amount of carbon dioxide.

123. Which of the following organisms is used in the production of alcoholic beverages?

- (1) *Penicillium notatum*
- (2) *Saccharomyces cerevisiae*
- (3) *Aspergillus niger*
- (4) *Clostridium butylicum*

124. BOD of wastewater is estimated by measuring the amount of:

- (1) carbon dioxide present.
- (2) biodegradable organic matter.
- (3) oxygen evolution.
- (4) oxygen consumption.

125. Which of the following statement is/are **correct**?

- (1) Toddy is made by fermenting sap from palm trees.
- (2) Microbes are also used to ferment fish, soyabean and bamboo-shoots.
- (3) Microbes are also used for making cheese.
- (4) All of these

126. Baculoviruses are excellent candidates for:

- (1) species-specific narrow spectrum pesticidal applications.
- (2) species-specific broad spectrum pesticidal applications.
- (3) species-specific narrow spectrum insecticidal applications.
- (4) species-specific broad spectrum insecticidal applications.

127. Conversion of milk to curd improves its nutritional value by increasing the amount of:

- (1) vitamin D.
- (2) vitamin A.
- (3) vitamin B_{12} .
- (4) vitamin E.

128. Match **list I** with **list II**.

List-I		List-II	
(A)	Blue green algae	(I)	Get rid of mosquitoes
(B)	IPM	(II)	Present in root ecosystems
(C)	<i>Trichoderma</i>	(III)	Integrated pest management
(D)	Dragonflies	(IV)	Add organic matter to soil

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-III, D-IV
- (2) A-II, B-III, C-IV, D-I
- (3) A-II, B-IV, C-I, D-III
- (4) A-IV, B-III, C-II, D-I

129. Choose the **incorrect** statement.

- (1) Pectinases are used for clarification of bottled juices.
- (2) Lipases are used in detergent formulations.
- (3) Streptokinase used as clot buster.
- (4) Proteases are used as immunosuppressive agent.

130. Which of the following can be used as a biocontrol agent in the treatment of plant disease?

- (1) *Chlorella*
- (2) *Anabaena*
- (3) *Lactobacillus*
- (4) *Trichoderma*

131. Cheese and yogurt are products of:

- (1) pasteurisation.
- (2) fermentation.
- (3) dehydration.
- (4) distillation.

132. In an angiospermic plant the embryo develops:

- (1) at micropylar end of embryo sac.
- (2) at chalazal end of embryo sac.
- (3) from nucellus only.
- (4) always prior to fertilization.

133. A true fruit is the one where:

- (1) only ovary of the flower develops into fruit.
- (2) ovary and calyx of the flower develops into fruit.
- (3) ovary, calyx and thalamus of the flower develops into fruit.
- (4) All floral whorls of the flower develops into fruit.

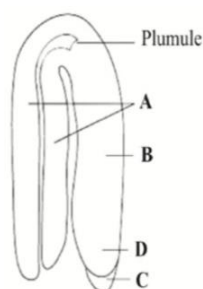
134. Identify the **correct** statement(s).

- A. Hybrid varieties of several of our food and vegetable crops are being extensively cultivated.
- B. Cultivation of hybrids has tremendously increased productivity.
- C. One of the problems of hybrids is that hybrid seeds have to be produced every year.
- D. Production of hybrid seeds is costly.

Choose the most appropriate answer from the options given below:

- (1) A only (2) A and B only
- (3) A and C only (4) A, B, C and D

135. Choose the option showing the **correct** labelling A, B, C and D in the given figure of a dicot embryo.



- (1) A - Hypocotyl; B - Cotyledons; C - Root cap; D-Radicle
- (2) A - Cotyledons; B - Hypocotyl; C - Root cap; D-Radicle
- (3) A - Cotyledons; B - Hypocotyl; C - Radicle; D-Root cap
- (4) A - Cotyledons; B - Radicle; C - Hypocotyl; D - Root cap.

SECTION - B

136. In some dicots, no distinct endosperm can be seen. Why?

- (1) The embryo has digested the endosperm.
- (2) The cotyledons have absorbed the endosperm.
- (3) The seeds never produced endosperm.
- (4) The endosperm has become the seed coat.

137. Which one of the following pairs of plant structures has haploid number of chromosomes?

- (1) Nucellus and antipodal cells
- (2) Egg nucleus and polar nuclei
- (3) Megaspore mother cell and antipodal cells
- (4) Egg cell and antipodal cells

138. Given below are two statements:

Statement I: The transformation of ovules into seeds and ovary into fruit proceeds simultaneously.

Statement II: The wall of the ovule develops into the wall of fruit.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

139. From among the situations given below, choose the one that prevents both autogamy and geitonogamy.

- (1) Monoecious plant bearing unisexual flowers
- (2) Dioecious plant bearing only male or female flowers
- (3) Monoecious plant with bisexual flowers
- (4) Dioecious plant with bisexual flowers

140. The male gametes of Rice plant have 12 chromosomes in their nucleus. The chromosome number in the female gamete and zygote will be, respectively?

- (1) 24, 12 (2) 12, 12
- (3) 12, 24 (4) 24, 36

141. Read the following statements regarding pollen.

- A. A large number of pollen products in the form of tablets and syrups are available in the market.
- B. Pollens have fascinating array of patterns and designs of pectin on outer wall.
- C. Pollens can be used as food supplement.
- D. Pollen grains are rich in nutrients.

Choose the most appropriate answer from the options given below.

- (1) A, C and D are correct only
- (2) A and C are correct only
- (3) B, C and D incorrect
- (4) D is correct only

142. Choose the **correct** option w.r.t. to the germ pore.

- (1) It allows growth of pollen tube.
- (2) It is present in stigma.
- (3) It is present in inner layer of the pollen grain.
- (4) It is made up of sporopollenin.

- 143.** Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Pollen mother cells (PMCs) give rise to four pollen grains.

Reason R: Each PMC undergoes meiosis.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false
- (2) A is false but R is true
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

- 144.** Adaptation shown by pollen grain by hydrophilous flower is:

- (1) hairy exine of pollen grains.
- (2) mucilage coat on pollen grains.
- (3) heavy weight pollen grains.
- (4) winged pollen grains.

- 145.** Mark the **correctly** matched.

- (1) *Vallisneria* – underwater pollination
- (2) *Zostera* – ribbon shape pollen
- (3) Maize – mucilaginous covering on pollen
- (4) *Amorphophallus* plant – wind pollinated

- 146.** Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Endosperm development precedes embryo development.

Reason R: Endosperm provides the nutrition of the developing embryo.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

- 147.** Which one of the following statements is **incorrect**?

- (1) Sometimes the generative cell divides during the growth of pollen tube in the stigma.
- (2) In some species, floral rewards are providing safe places to lay eggs.
- (3) The portion of embryonal axis above the level of cotyledons is the hypocotyl.
- (4) Scutellum is lateral to the embryonal axis in monocots.

- 148.** Match **List-I** with **List-II**:

List-I		List-II	
(A)	Polyembryony	(I)	Inactive state of embryo
(B)	Apomixis	(II)	Thalamus also contributes to fruit formation
(C)	Dormancy	(III)	Occurrence of more than one embryo in a seed
(D)	False fruit	(IV)	Seed formation without fertilisation

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-IV, D-III
- (2) A-IV, B-III, C-I, D-II
- (3) A-III, B-IV, C-I, D-II
- (4) A-I, B-II, C-IV, D-III

- 149.** To ensure that only the desired pollens fall on the stigma in artificial hybridization process:

- A. the female flower buds of plant producing unisexual flowers need not be bagged.
- B. there is no need to emasculate unisexual flowers of selected female parent.
- C. emasculated flowers are to be bagged immediately after cross pollination.
- D. emasculated flowers are to be bagged after removal of anthers bisexual flowers, showing protogyny are never selected for cross.
- E. bisexual flowers, showing protogyny are never selected for cross.

Choose the correct answer from the options given below:

- (1) A, D and E only
- (2) A, B and C only
- (3) B, C and D only
- (4) B, C and E only

- 150.** Match **List-I** with **List-II**:

List-I		List-II	
(A)	Tassels of a corn cob	(I)	Insect pollination
(B)	Beetles secrete foul odours	(II)	Provide safe place to lay eggs for insect
(C)	<i>Yucca</i>	(III)	Wind pollination
(D)	<i>Amorphophallus</i>	(IV)	Get pollinated by moth

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
- (2) A-I, B-III, C-II, D-IV
- (3) A-IV, B-I, C-III, D-II
- (4) A-III, B-I, C-IV, D-II

SECTION-(IV) ZOOLOGY

SECTION-A

151. Which of the following is **not** the reason of increase of population in India?
- (1) Decrease in infant mortality rate.
 - (2) Decrease number of people reaching reproductive age.
 - (3) Decline in death rate.
 - (4) Decline in maternal mortality rate.
152. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Both male and female can be infertile.
Reason R: Infertile couples can have children with the help of ARTs.
In the light of the above statements, choose the **correct** answer from the options given below.
- (1) A is true but R is false.
 - (2) A is false but R is true.
 - (3) Both A and R are true and R is the correct explanation of A.
 - (4) Both A and R are true but R is not the correct explanation of A.
153. Early symptoms of most STDs are:
- (i) itching.
 - (ii) fluid discharge.
 - (iii) slight pain.
 - (iv) swelling.
- (1) (i), (ii) and (iv) only
 - (2) (i), (ii) and (iii) only
 - (3) (i), (iii) and (iv) only
 - (4) (i), (ii), (iii) and (iv)
154. What is the primary goal of implementing various action plans for reproductive health?
- (1) To promote population growth.
 - (2) To provide reproductive assistance and care.
 - (3) To ensure high child mortality rates.
 - (4) To restrict access to contraceptives
155. Which of the following statements is **incorrect** regarding MTP?
- (1) Intentional or voluntary termination of pregnancy before full term is called medical termination of pregnancy.
 - (2) MTPs are safe during the third trimester.
 - (3) Majority of the MTPs are performed illegally by unqualified quacks.
 - (4) Government of India legalised MTP in 1971.

156. Refer to the following diagram and choose the **correct** option.



- (1) It is a male condom which is used to cover the penis just before the coitus to prevent the entry of ejaculated semen into the female reproductive tract.
 - (2) It is a female condom which is used to cover the cervix and vagina just before the coitus.
 - (3) It is a condom which is used to cover oviduct in female.
 - (4) It is one type of IUDs which makes the uterus unsuitable for implantation and cervix hostile to the sperms.
157. Given below are two statements:
Statement I: In vasectomy, a small part of the vas deferens of male is tied or cut.
Statement II: Genital herpes is completely curable if detected early and treated properly.
In the light of the above statements, choose the most appropriate answer from the options given below.
- (1) Statement I is correct but Statement II is incorrect.
 - (2) Statement I is incorrect but Statement II is correct.
 - (3) Both Statement I and Statement II are correct.
 - (4) Both Statement I and Statement II are incorrect.
158. A childless couple can be assisted to have a child through a technique called GIFT. The full form of this technique is:
- (1) gamete inseminated fallopian transfer.
 - (2) gamete intrafallopian transfer.
 - (3) gamete internal fertilisation & transfer.
 - (4) germ cell internal fallopian transfer.
159. Which of the following is the primary function of the contraceptive 'Saheli'?
- (1) To treat infertility.
 - (2) To prevent sexually transmitted diseases (STDs).
 - (3) To act as a hormonal contraceptive for women.
 - (4) To aid in post-natal care.

160. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The female gamete is produced at the time of gestation.

Reason R: Gonadotropin releasing hormone controls the process of oogenesis. In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

161. The extinct human ancestor who ate only fruits and hunted with stone weapons was:

- (1) *Ramapithecus*.
- (2) *Australopithecines*.
- (3) *Dryopithecus*.
- (4) *Homo habilis*.

162. Choose the **odd** one out w.r.t. vestigial structures in humans.

- (1) Vermiform appendix
- (2) Wisdom teeth
- (3) Body hair
- (4) Epiglottis

163. Who proposed embryological support for evolution?

- (1) Charles Darwin
- (2) Karl Ernst von Baer
- (3) Ernst Haeckel
- (4) Alfred Wallace

164. Who disapproved the theory of spontaneous generation?

- (1) Urey and Miller
- (2) Oparin
- (3) Louis pasteur
- (4) Aristotle

165. Match **List-I** with **List-II**.

List-I Contraceptive Method		List-II Device	
(A)	Barrier method	(I)	Cervical cap
(B)	Natural method	(II)	Copper-T
(C)	IUDs	(III)	Implant
(D)	Subcutaneous method	(IV)	Coitus interruptus

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
- (2) A-I, B-IV, C-III, D-II
- (3) A-I, B-IV, C-II, D-III
- (4) A-IV, B-I, C-II, D-III

166. Which of the following organisms are thought to be evolved into first amphibians?

- (1) *Petromyzon*
- (2) *Lamprey*
- (3) *Coelacanth*
- (4) *Ascidia*

167. Amniocentesis is the procedure to detect:

- (1) chromosomal pattern of the foetus by obtaining amniotic fluid.
- (2) amount of chorionic fluid from developing embryo.
- (3) chromosomal pattern of child after childbirth.
- (4) chromosomal pattern before fertilisation.

168. Which of the following is **not** an IUD?

- (1) Vaults
- (2) CuT
- (3) Multiload 375
- (4) Progestasert

169. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Sterilisation is a procedure that renders an individual incapable of further reproduction.

Reason R: Vasectomy is a principal method of sterilisation in females.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

170. From his experiments, S.L. Miller produced amino acids by mixing the following in a closed flask:

- (1) CH₃, H₂, NH₄ and water vapor at 800°C
- (2) CH₄, H₂, NH₃ and water vapor at 600°C
- (3) CH₃, H₂, NH₃ and water vapor at 600°C
- (4) CH₄, H₂, NH₃ and water vapor at 800°C

171. At a particular locus, frequency of 'A' allele is 0.3 and that of 'a' is 0.7, what will be the frequency of heterozygotes in random mating population at equilibrium?

- (1) 0.48
- (2) 0.21
- (3) 0.42
- (4) 0.9

172. Flying squirrel and flying phalanger show:

- (1) divergent evolution.
- (2) convergent evolution.
- (3) adaptive radiation.
- (4) both (1) and (2)

173. A gene pool is:

- (1) total genes present in a cell only.
- (2) total genes present in the body only.
- (3) total dominant genes only.
- (4) total genes and their alleles in a population.

174. The cell division in secondary oocyte is suspended at:

- (1) Anaphase II.
- (2) Metaphase II.
- (3) Prophase I.
- (4) Telophase II.

175. Select the **correct** option.

- (1) *Ramapithecus* was more ape like.
- (2) *Australopithecines* probably lived in East African grasslands.
- (3) *Homo erectus* had a cranial capacity of 700 cc.
- (4) *Homo habilis* probably eat meat.

176. Change in the female secondary sex organs are controlled by _____ of the ovary in the proliferative phase of the menstrual cycle whereas the changes in the secretory phase are controlled by the hormone.

- (1) estrogen
- (2) progesterone
- (3) testosterone
- (4) prolactin

177. Which of the following statements is **incorrect**?

- (1) The secretions of the acrosome help the sperm to enter into the cytoplasm of the ovum.
- (2) Oviduct is 10-12 cm in length.
- (3) LH acts at the Leydig cells.
- (4) Prostate gland is a paired gland in the male reproductive structure.

178. How many sperms are formed from a secondary spermatocyte?

- (1) Four
- (2) Eight
- (3) Two
- (4) One

179. The glandular tissue of each breast is divided into _____ mammary lobes.

Choose the option which fills the blank **correctly**.

- (1) 10 – 12
- (2) 12 – 15
- (3) 15 – 20
- (4) 20 – 50

180. In Hardy-Weinberg principle, the frequency of AA individuals in a population is represented as:

- (1) p^2 .
- (2) $2pq$.
- (3) pq .
- (4) q^2 .

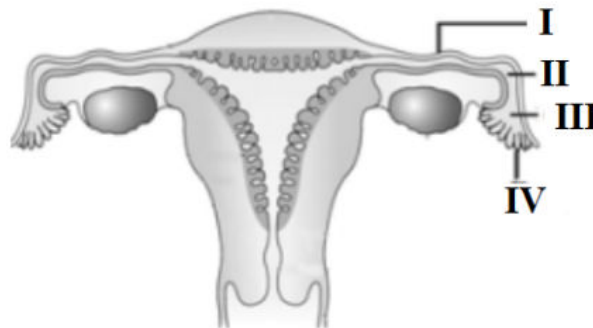
181. Which of the following is **not** a placental hormone?

- (1) Progesterone
- (2) Human Chorionic Gonadotropin (hCG)
- (3) Estrogen
- (4) Melatonin

182. Which of the following is **incorrect** match?

- (1) Primary follicles at puberty - 60000 - 80000
- (2) Antrum - Fluid filled cavity
- (3) Child birth - Colostrum
- (4) Daughter cells – Blastomeres

183. The given diagram shows the diagrammatic sectional view of female reproductive system with few structures marked as I, II, III and IV. Select the option which shows the correct identification of the structure with its characteristics.



- (1) **I:** Infundibulum, funnel shaped structure surrounded by finger like projection.
- (2) **II:** Ampulla, wider part of oviduct where fertilisation occurs.
- (3) **III:** Isthmus, it has a narrow lumen and joins with uterus
- (4) **IV:** Fimbriae, it collects ovum before ovulation.

184. How many of the following statements regarding implantation are **correct**?

- A. It refers to the attachment of blastocyst to the uterine wall.
- B. It occurs immediately before the fertilisation.
- C. It occurs under the influence of estrogen and progesterone.
- D. It results in increase in the size of uterus and thickening of perimetrium.

- (1) One (2) Two
- (3) Three (4) Four

185. Given below are two statements.

Statement-I: Labia minora extend down from the mons pubis.

Statement-II: The opening of the vagina is often covered partially by the hymen.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

SECTION - B

186. According to Lamarckism, long necked giraffes evolved because:

- (1) nature selected only long necked ones.
- (2) humans preferred only long necked ones.
- (3) short necks suddenly changed into long necks.
- (4) of stretching of necks over many generations by short necked ones.

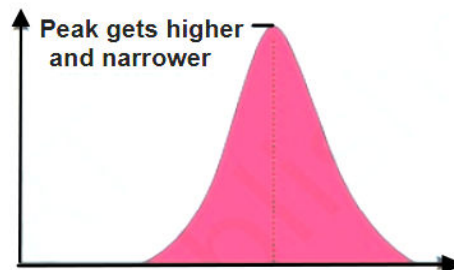
187. Match the **List-I** with **List-II**.

List-I		List-II	
(A)	Gametogenesis	(I)	Delivery of the baby.
(B)	Insemination	(II)	Release of secondary oocyte from ovary
(C)	Parturition	(III)	Formation of gametes.
(D)	Ovulation	(IV)	Transfer of sperms into female genital tract.

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
- (2) A-I, B-II, C-III, D-IV
- (3) A-I, B-III, C-II, D-IV
- (4) A-III, B-IV, C-II, D-I

188. Refer to the given below diagram.



Which of the following type of natural selection is depicted in the above diagram?

- (1) Stabilising
- (2) Directional
- (3) Disruptive
- (4) All of these

189. The meiotic division of the secondary oocyte is completed:

- (1) at the time of copulation.
- (2) after zygote formation.
- (3) at the time of fusion of a sperm with an ovum.
- (4) prior to ovulation.

190. The phase of embryonic development during pregnancy is _____.

Choose the option which fills the blank **correctly**.

- (1) gestation. (2) insemination.
- (3) implantation. (4) parturition.

191. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Origin of universe	(I)	4.5 bya
(B)	Origin of earth	(II)	20 bya
(C)	Appearance of life after the formation of earth	(III)	2000 mya
(D)	First cellular form of life	(IV)	500 mya

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-III, D-IV
- (2) A-II, B-I, C-IV, D-III
- (3) A-III, B-I, C-II, D-IV
- (4) A-III, B-I, C-IV, D-II

192. Given below are two statements:
Statement I: The atmosphere of primitive earth was reducing in nature.
Statement II: *Tyrannosaurus rex* had huge fearsome dagger-like teeth.
In the light of the above statements, choose the most appropriate answer from the options given below.
- (1) Statement I is correct but Statement II is incorrect.
 - (2) Statement I is incorrect but Statement II is correct.
 - (3) Both Statement I and Statement II are correct.
 - (4) Both Statement I and Statement II are incorrect.
193. About which day in a normal human menstrual cycle does rapid secretion of LH (popularly called LH surge) normally occurs?
- (1) 14th day
 - (2) 20th day
 - (3) 5th day
 - (4) 11th day
194. Which of the following statements would Darwin agree with?
- (1) Individuals within a population vary in the characteristics they possess
 - (2) Evolution is best viewed as a purposeful and directed change over time
 - (3) Natural selection is the mechanism by which biological evolution takes place
 - (4) All of these
195. By the end of the second month, which of the following features are developed in the foetus?
- (1) Fine hair and eyelashes
 - (2) Limbs and digits
 - (3) External genital organs
 - (4) Eye-lids separate
196. Read the following statements (A-E).
- A. Spermicidal creams, jellies and foams are usually used along with barriers to increase their contraceptive efficiency.
 - B. LNG-20 is a type of hormone releasing IUD.
 - C. Cervical canal along with oviduct forms the birth canal.
 - D. Ovaries are the primary female sex organ.
 - E. Lactational amenorrhea is the absence of menstruation in lactating mothers.

How many of the above statements is/are **incorrect**?

- (1) One
- (2) Two
- (3) Three
- (4) Four

197. Which of the following statements is **incorrect**?
- (1) Menstruation does not take place if the ovum is fertilised.
 - (2) The milk produced during initial few days of lactation is called colostrum.
 - (3) The corpus luteum secretes large amount of oxytocin which is essential for maintenance of myometrium.
 - (4) The seminal plasma along with the sperm constitute semen.
198. Use of which of the following contraceptive device has increased in recent years due to its additional benefit of protecting the user from contracting STDs and AIDS?
- (1) Vasectomy
 - (2) Coitus interruptus
 - (3) Condoms
 - (4) Intra Uterine Devices
199. Which of the following contributes in the formation of seminal plasma?
- (a) Seminal vesicle
 - (b) Sertoli cells
 - (c) Prostate gland
 - (d) Vas deferens
 - (e) Bulbourethral gland
- (1) a, c, and d only
 - (2) a, b, and c only
 - (3) a, c and e only
 - (4) c and d only
200. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Darwin's finches are a classical example of natural selection.
Reason R: Darwin explained that the difference in size and shape of the beaks of the species was a result of the adaptation of the species to different food habits.
In the light of the above statements, choose the **correct** answer from the options given below.
- (1) A is true but R is false.
 - (2) A is false but R is true.
 - (3) Both A and R are true and R is the correct explanation of A.
 - (4) Both A and R are true but R is not the correct explanation of A.

